



FLEET & SHORE PERSONNEL TRAINING

Environmental and Energy Management

The Company Environmental and Energy Management Plan — how energy performance is planned, controlled, measured and improved on board and in the office.

ISO 50001

EU MRV Regulation

MARPOL Annex VI Reg. 22A

TMSA3 Element 10

What we will cover

01

The Energy Management Plan

Scope, purpose and structure

02

Management and Responsibility

Commitment, teams and the energy policy

03

Energy Planning

Review, requirements, baseline and targets

04

Implementation and Operation

Competence, awareness and documentation

05

Control — Voyage and Navigation

Trim, speed, routeing and arrival

06

Control — Machinery and Power

Main engine, auxiliaries and boilers

07

Control — Hull, Propeller, Ballast

Condition, devices and ballast practice

08

Checking and Monitoring

Measurement, audit and records

09

Management Review

Inputs, outputs and improvement



SECTION 01

The Energy Management Plan

The Company has established, documented and implements the requirements defined within this Energy Management Plan as an Energy Management System.

What the Energy Management Plan is

The Plan sets out the requirements the Company operates as its Energy Management System, and the Company aims to maintain and improve that system.



Established and documented

The requirements are written down and implemented as a management system, not left to custom and practice.



Built on ISO 50001

The Plan has been prepared in accordance with ISO 50001, the international standard for energy management.



Maintained and improved

The Company aims to maintain the system and to improve it continually, rather than treat it as a fixed document.



Aimed at performance

The purpose is continual improvement of energy performance as well as of the management system itself.

What the Plan covers

The Energy Management Plan covers shipboard operations and office energy consumption alike.



Shipboard operations

The fleet

- Cargo related operations and navigation are the activities reviewed for energy performance
- Ship types, specifications and engines including auxiliaries are detailed within the related MRV Plans and SEEMPs
- Requirements are implemented by responsible vessel staff



Office energy consumption

Ashore

- Use of electricity and natural gas is the consumption reviewed
- Requirements are implemented by responsible office staff
- The nature and scale of the organisation energy use and consumption set the boundaries of the system

THE CORE IDEA

Energy management is a structured process, not an occasional effort

It is a structured process running through the organisation which seeks to optimise energy use, and whose definition and practice is codified through ISO 50001.

Structured

Measured

Reviewed

Improved

From requirement to improvement

Energy Management Plan requirements are implemented by responsible vessel staff and office staff, and the outcomes are monitored and reviewed by assigned persons.



Requirements are assigned

Each requirement of the Plan has responsible vessel staff or office staff attached to it.



Outcomes are monitored

Assigned persons monitor the outcomes of the measures put in place, on board and ashore.



Outcomes are reviewed

Results are reviewed against the requirements of ISO 50001 at planned intervals.



Performance improves continually

The aim of the cycle is continual improvement of energy performance and of the Energy Management System.



Two systems, one plan

Shipboard and office energy use are managed under the same Plan, with the same policy, the same review cycle and the same reporting rhythm. Only the measured quantities differ.

The five parts of the Energy Management System

The Plan follows the classic management-system cycle. Each part of this course maps onto one stage.





SECTION 02

Management and Responsibility

Top management commits to support the Energy Management System and to improve its effectiveness continually.

02

The tools top management uses

Top management of the Company commits to support the Energy Management System and to continually improve its effectiveness. These are the tools for that aim.



An energy policy

The Energy Policy is defined, established, implemented and maintained as set out in this chapter of the Plan.



An Energy Efficiency Officer

The Operational Manager is appointed as Energy Efficiency Officer, as defined in M1 Main Manual Chapter 3.1.39.



A team on every vessel

Energy Management Teams are assigned for each vessel from the ranks of Master, Chief Officer and Chief Engineer.



A team in the office

The Human Resources Manager is assigned as responsible within the Office Energy Management Team together with the Operational Manager.

The Energy Management Teams

Team	Members	Responsible for
Management representative	Operational Manager, appointed as Energy Efficiency Officer under M1 Main Manual Chapter 3.1.39	Establishing, implementing, maintaining and continually improving the Energy Management System
Office Energy Management Team	Operational Manager and Human Resources Manager	Office energy consumption — electricity and natural gas
Vessel Energy Management Team	Master, Chief Officer and Chief Engineer, on each vessel	Shipboard operations — cargo related operations and navigation

The Energy Efficiency Officer was appointed considering the skills and competence required for the role.

What the Company provides

The Company provides the resources needed to establish, implement, maintain and improve the Energy Management System and the resulting energy performance.



Human resources

People assigned to energy management duties on board and in the office.



Specialised skills

The competence needed to run the system and to judge energy performance.



Technology

Systems, software and equipment that make measurement and control possible.



Financial sources

Funding for the measures and improvements the system identifies.

What the Company does with those resources



Communicates the importance of energy management

To all people in the organisation during pre-joining trainings, in-house trainings and Company-run seminars, and by quarterly energy review reports sent by e-mail.



Reviews the performance indicators

The Company reviews and ensures that Energy Performance Indicators are appropriate to the organisation.



Plans for the long term

The Company makes long-term planning for energy performance rather than reacting voyage by voyage.



Measures and reports quarterly

The Company ensures that results are measured and reported at quarterly intervals.



Conducts management reviews

The Company conducts management reviews of the Energy Management System.

Responsibility and authority of the Energy Efficiency Officer

Irrespective of other responsibilities, the Energy Efficiency Officer has the responsibility and authority to:

- Ensure the Energy Management System is established, implemented, maintained and continually improved in accordance with ISO 50001
- Identify persons, authorised by an appropriate level of management, to work with the management representative in support of energy management activities
- Report to top management on energy performance monthly, with the Energy Efficiency Management Report
- Report to top management on the performance of the Energy Management System with quarterly management review meeting reports, prepared after review meetings with key staff of the Company
- Ensure that the planning of energy management activities is designed to support the energy policy, considering the outcomes of vessel visits and office audits
- Define and communicate responsibilities and authorities to facilitate effective energy management, whenever necessary beyond those already defined in the Plan
- Determine the criteria and methods needed to ensure that operation and control of the Energy Management System are effective
- Promote awareness of the energy policy and objectives at all levels of the organisation



One named person

The role sits with the Operational Manager. Naming a single management representative is what makes the system auditable — there is always someone accountable for its condition.

When energy performance is reported

Report or activity	Frequency	Audience
Energy Efficiency Management Report	Monthly	Top management — energy performance
Management review meeting report	Quarterly	Top management — performance of the Energy Management System
Energy review report by e-mail	Quarterly	All people in the organisation
Measurement and reporting of results	Quarterly	Top management and key staff
Management review meeting	Annual	Top management and key staff of the Company

ENERGY CONSERVATION POLICY

Use every form of energy in a cost effective and efficient manner

Glory Ship Management is committed to utilising various forms of energy in a cost effective and efficient manner in order to conserve energy resources and achieve energy performance improvement for a cleaner environment.

Shipboard operations

Office consumption

A cleaner environment

What the policy commits the Company to

The policy covers shipboard operations and office energy consumption, considering the nature and scale of the organisation energy use and consumption.



Minimise use and cost

Minimise the use of energy and its purchased cost by continuously measuring and controlling the energy index and its cost.



Stay proportionate

Keep the Energy Management Plan and Policy appropriate to the nature and scale of the organisation energy use and consumption.



Promote an awareness culture

Promote an energy awareness culture in the organisation and amongst employees.



Improve continually

Aim at continual improvement in energy performance rather than one-off savings.



Make resources available

Ensure the availability of information and of the resources necessary to achieve objectives and targets.

Audit, compliance and procurement



Periodic energy audits

Ensure periodic energy audits are carried out, and that identified improvements are implemented.



Legal compliance

Comply with applicable legal and other requirements to which the organisation subscribes for energy use, consumption and efficiency.



A framework for targets

Provide the framework for setting and reviewing energy objectives and targets.



Efficient purchasing

Support the purchase of energy-efficient products and services.



Design for performance

Support design for energy performance improvement.



SECTION 03

Energy Planning

A documented planning process, consistent with the energy policy, that leads to activities which continually improve energy performance.

03

Inputs, review and outputs

The Company conducted and documented an energy planning process that reviews the activities which can affect energy performance.



Planning inputs

- Past and present energy uses
- Relevant variables affecting significant energy use
- Performance data



Energy review

- A — analyse energy use and consumption
- B — identify areas of significant energy use
- C — identify opportunities for improving energy performance



Planning outputs

- Energy baseline
- Energy Performance Indicators
- Objectives
- Targets
- Action plans

The activities that can affect energy performance

Energy planning involved a review of the organisation activities that can affect energy performance.



Shipboard operations

Cargo related operations and navigation were reviewed. These are the activities where fuel is consumed in quantity and where the crew has real influence.



Office energy consumption

Use of electricity and natural gas was reviewed. Both are followed through the official bills received by the office.

What the Company has chosen to comply with

Presently there is no legal requirement for office and shipboard Energy Management Systems. The Company has nevertheless identified the requirements below to implement, in order to comply with ISO 50001.



EU MRV Shipping Regulation

Monitoring, reporting and verification of CO2 emissions based on the fuel consumption of ships.



IMO Fuel Consumption Data Collection System

Mandatory collection and reporting of ship fuel oil consumption data to an IMO database.



TMSA3 Element 10

An environmental management plan that includes energy efficiency and fuel management.



Voluntary today, mandatory tomorrow

The Company treats these as requirements now, so that no change of practice is needed when they become binding. Once a new regulation or requirement is identified, the Company Management of Change Procedure is followed for compliance.

EU MRV Shipping Regulation

In order to reduce emissions from shipping, the European Union introduced the shipping monitoring, reporting and verification regulation.



What it gathers

Data on CO2 emissions, based on the fuel consumption of ships.



In force July 2015

The MRV regulation entered into force in July 2015.



Data from 01.01.2018

Data collection started with effect from 1 January 2018.



Company MRV Plans

The Company has prepared MRV Plans for applicable fleet vessels and implements the requirements accordingly.

REQUIREMENT 2

IMO Fuel Consumption Data Collection System

The IMO has adopted a global approach — a mandatory fuel consumption data collection system for international shipping.

5,000

Gross tonnage and above — the ships required to collect and report data

2019

The year from which data is reported to an IMO database

22A

The new regulation added to MARPOL Annex VI, chapter 4



How it was adopted — As an amendment to chapter 4 of MARPOL Annex VI, adding a new regulation on the collection and reporting of ship fuel oil consumption data.



Why it exists — The aim is to reduce emissions, and reduction of emissions will come with energy saving.



Company position — The Company will comply with the IMO Fuel Consumption Data Collection System once it is in force.

TMSA3 Element 10

TMSA3 requires companies to have an environmental management plan which includes energy efficiency and fuel management.



A plan is required

The environmental management plan must address the efficient use of energy.



Actions, not intentions

The plan includes actions to improve environmental performance, not statements of aspiration alone.



Optimisation is the aim

The Company seeks to optimise vessel energy efficiency across the fleet.

How requirements are identified and reviewed



Reviewed annually

Legal requirements and other requirements are reviewed during the annual management review meetings.



Legal Requirements Implementation Table

For office legal requirements, the Company uses the Legal Requirements Implementation Table.



Classification society data

Data from classification societies is used to identify legal and other requirements.



regs4ships and Lebib Yalkin

The regs4ships software and the Lebib Yalkin webpage are also used to identify requirements.

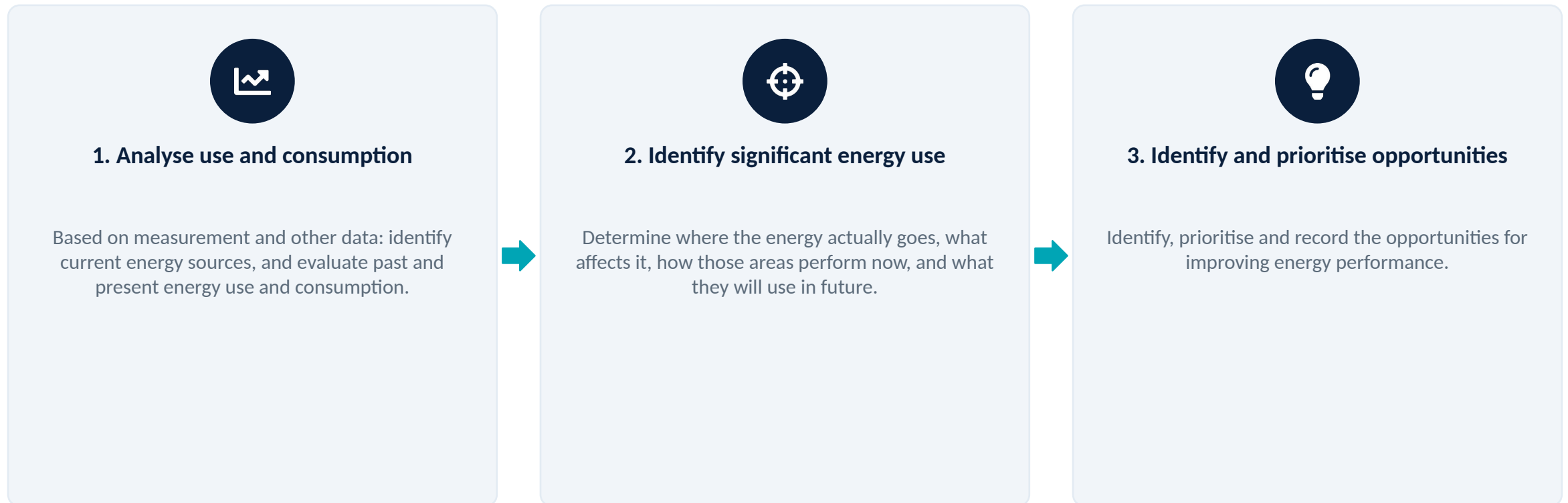


Management of Change

Once a new regulation or requirement is identified, the Company Management of Change Procedure is followed for compliance.

The three stages of the energy review

The energy review is conducted quarterly, during the annual management review meetings, and also in response to major changes in facilities, equipment, systems or processes.



What identifying significant energy use means

Based on the analysis of energy use and consumption, the areas of significant energy use are identified as follows.



Facilities and people

Identify the facilities, equipment, systems, processes and personnel working for or on behalf of the organisation that significantly affect energy use and consumption.



Relevant variables

Identify the other relevant variables that affect those significant energy uses.



Current performance

Determine the current energy performance of the facilities, equipment, systems and processes related to the identified significant energy uses.



Future consumption

Estimate future energy use and consumption on the basis of that performance.

The line performance is measured against

The Company established the energy baseline using the information in the initial energy review, considering a data period suitable to Company use and consumption.

2017

The year-end data used as the energy baseline

Annual

How often the baseline is reviewed and performance measured against it



Indicators no longer represent use — The baseline is adjusted where Energy Performance Indicators no longer reflect organisational energy use and consumption.



Major changes occur — The baseline is adjusted where there have been major changes to the process, operational patterns or energy systems.



A pre-determined method applies — The baseline is also adjusted according to a pre-determined method. The energy baselines are maintained within App 21 3 - Company Targets.

Energy objectives, targets and action plans

The Company established, implemented and maintains documented energy objectives and targets as defined in the Environmental Management Programme.



Time frames are fixed

Time frames are established for the achievement of the objectives and targets.



Everything stays consistent

The objectives and targets are consistent with the energy policy, and the targets are consistent with the objectives.



Requirements are taken into account

Legal requirements and other requirements, significant energy uses and improvement opportunities identified in the energy review are all considered.



Reality is taken into account

Financial, operational and business conditions, technological options and the views of interested parties are also considered.

What every action plan must contain

The Company established, implemented and maintains action plans for achieving its objectives and targets as necessary. Any action plan established is required to include the following.



Designation of responsibility

A named person or role accountable for delivering the action.



Means and time frame

The means and time frame by which individual targets are to be achieved.



How improvement is verified

A statement of the method by which an improvement in energy performance shall be verified.



How results are verified

A statement of the method of verifying the results achieved.



SECTION 04

Implementation and Operation

The Company uses action plans and other outputs of the planning process for implementation and operation.

04

Competence for significant energy uses

The Company ensures that all crew working on board, or in the office for Glory Ship Management, whose work relates to significant energy uses, are competent.



Appropriate education

Competence is established on the basis of appropriate education for the duties held.



Training

Training relevant to the control of significant energy uses and to the operation of the Energy Management System.



Skills

Practical skills that allow the operational controls to be applied correctly on board and ashore.



Experience

Experience in the operations that consume energy in quantity.



Training needs are identified

The Company identifies the training needs associated with the control of its significant energy uses and the operation of its Energy Management System.

Who is trained, and when

The Company provides the necessary training to its crew members and to its office staff.



Office staff

On the office plan

- Each member of office staff should be trained in the content and requirements of the Office Environmental and Energy Management Plan
- Training is delivered as required by the Office Environmental and Energy Plan requirements



Crew members

Before joining

- Energy conservation training should be arranged for each crew member prior to joining their assigned vessel
- Awareness is also raised during pre-joining trainings, in-house trainings and Company-run seminars

What every crew member must be aware of

The Company ensures all crew members are aware of the following.



Why conformity matters

The importance of conformity with the energy policy, the procedures and the requirements of the Energy Management System.



Their own part in it

Their roles, responsibilities and authorities in achieving the requirements of the Energy Management System.



The benefits

The benefits that improved energy performance brings to the ship, the Company and the environment.



The impact of behaviour

The actual or potential impact of their activities on energy use and consumption, how their behaviour contributes to objectives and targets, and the potential consequences of departing from specified procedures.

How energy performance is communicated

Internally and regularly



The Company communicates internally about its energy performance and Energy Management System regularly, using the annual management review meeting reports.

Anyone may contribute



Any person working for, or on behalf of, the Company can make comments or suggest improvements to the Energy Management System.

No external communication



The Company does not communicate externally about its energy policy, its Energy Management System or its energy performance.



Suggestions are wanted

Improvement ideas from seafarers carry weight because they come from the people operating the machinery. Raise them through the ship or through the office rather than keeping them to yourself.

What the Energy Management System documentation includes

In addition to the Plan itself, the Energy Management System includes the following documentation.

- The scope and boundaries of the Energy Management System, as defined within this chapter
- The energy policy, as defined within this chapter
- The energy objectives, targets and action plans, as defined within this chapter
- The documents, including records, as defined within this chapter
- Other documents, if determined by the organisation to be necessary



Records prove the system works

Documentation is what turns intentions into an auditable system. Records must remain legible, identifiable and traceable to the relevant activity.



SECTION 05

Control — Voyage and Navigation

The Company identifies and plans the operations and maintenance activities related to its significant energy uses, so that they are carried out under specified conditions.

05

How operational control is exercised

The operations and maintenance activities related to significant energy uses are planned so that they are consistent with the energy policy, objectives, targets and action plans.



Criteria are set

Criteria are established and set for the effective operation and maintenance of significant energy uses, where their absence could lead to a significant deviation from effective energy performance.



Criteria are applied

Facilities, processes, systems and equipment are operated and maintained in accordance with those operational criteria.



Criteria are communicated

The operational controls are appropriately communicated to personnel working for, or on behalf of, the Company.

WHY THIS MATTERS

Reduction of emissions will come with energy saving

Every operational control in this section exists for the same reason: fuel that is not burned is money that is not spent and emissions that are not released. The two objectives cannot be separated.

Less fuel

Lower cost

Fewer emissions

Optimisation of vessel trim

Indicated resistance varies according to the structure of the hull. A theoretical study is executed to determine the most appropriate trim state for each ship, and is provided for all our ships.



Use the trim program before departure

Use the program with the estimated departure drafts before the ship leaves on her voyage, together with estimated velocity and air current conditions, to determine the most convenient trim situation. The voyage is then executed in that trim.



Avoid load losses

The operation is carried out between ballasting and the start and end of the voyage, so far as possible, in order not to lead to load losses.



Do not create an overload

While applying the trim, an overload situation shall not be created, and proper attention shall be given to providing the required conditions where canal navigation is possible.



Report the trim to the office

Guidance for optimisation of vessel trim, where practical, is provided by the operational department to fleet vessels. Trim is reported to the office with noon reports for office control.

Speed optimisation

Speed optimisation can be counted as one of the most significant contributors to energy saving.



Subject to voyage conditions

Speed optimisation is to be conducted subject to voyage conditions and as per instructions received from the office.



ECL 12 for slow steaming

If slow steaming operations are conducted on board Suezmax and Capesize vessels, the requirements of ECL 12 Slow Steaming Control Check List are to be followed.



Verified on board

Compliance is verified during vessel visits such as audits and inspections, using information received from the operations department.

Weather routeing

Planning the voyage with regard to the weather reports can be counted as one of the significant contributors to energy saving.



Guidance is arranged where needed

Where a need is identified, such as ocean passage voyages, the operational department arranges weather routeing guidance to fleet vessels through service companies.



Agreed before departure

Before the voyage begins, the planned voyage is discussed and agreed in order to optimise it.



Verified on board

Compliance with the requirements of this section is verified during vessel visits such as audits and inspections.

Voyage optimisation

Masters should optimise route planning in consultation with the office, to avoid high storm or wave frequency and to maximise calm sea state during ocean passages.

- Take into consideration the effects of ocean current and tides
- Take into consideration the effects of weather systems
- Take into consideration crew safety and comfort, based on the trade and the route
- Voyage routes can be charted using Rhumb Lines and / or the Great Circle methodology
- Route planning is carried out in consultation with the office, not in isolation on board
- Compliance with the requirements of this section is verified during vessel visits such as audits and inspections

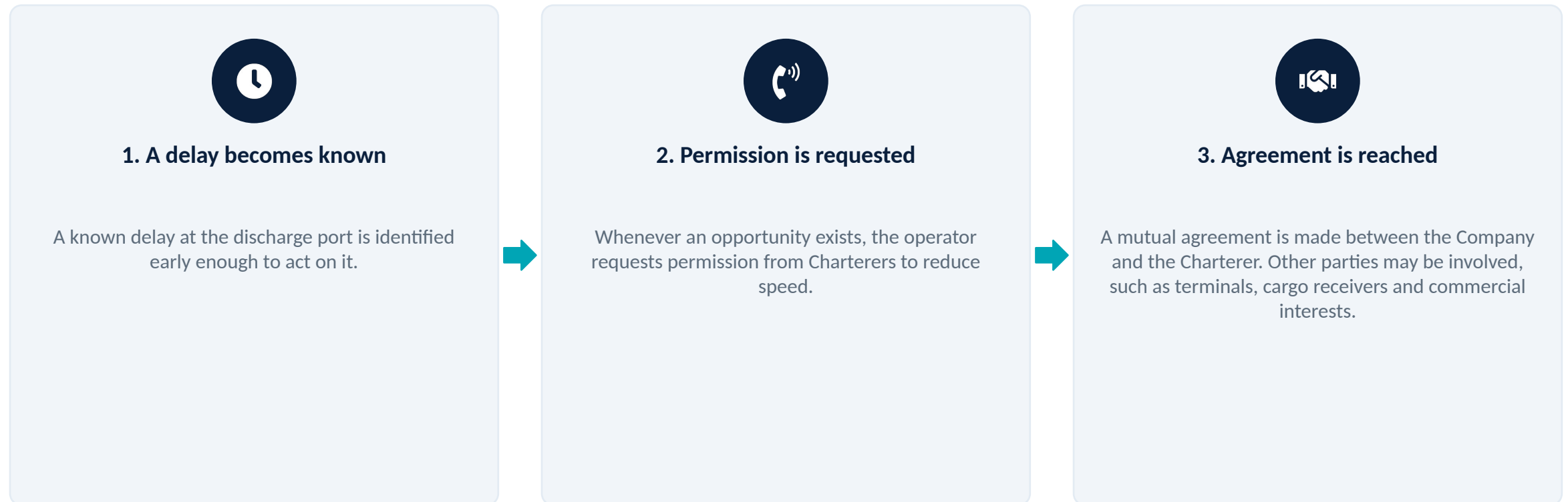


Responsibilities

- Office — Operational Manager
- Vessel — Master
- Verified during vessel visits, audits and inspections

Virtual arrival

Good early communication with the next port should be an aim of the office and of Masters, giving maximum notice of berth availability and allowing optimum speed where port operational procedures support this approach.



Auto pilot controls

There have been large improvements in automated heading and steering control systems. Originally developed to make the bridge team more effective, modern autopilots achieve much more.



What it delivers

Fuel saved off track

- An integrated navigation and command system can achieve significant fuel savings simply by reducing the distance sailed off track
- The principle is simple — better course control through less frequent and smaller corrections
- Smaller corrections minimise losses due to rudder resistance



Where it does not apply

Know the limits

- During approaches to ports and pilot stations the autopilot cannot always be used efficiently, as the rudder has to respond quickly to given commands
- At certain stages of the voyage it may have to be deactivated or very carefully adjusted, for example in heavy weather and on approaches to ports

Keeping the autopilot working for you



Adjust as the manufacturer instructs

All autopilot adjustments should be done in accordance with the manufacturer instructions.



Verify rudder movement

Movement of the rudder must be monitored during calm weather for verification that the settings are correct.



Take the software upgrades

Upgrades of the autopilot should be carried out whenever new software is available.



Verified on board

Compliance with the requirements of this section is verified during vessel visits such as audits and inspections.

Who is responsible for each voyage control

Operational control	Office	Vessel
Optimisation of vessel trim	Operational Manager	Master
Speed optimisation	Operational Manager and Superintendents	Master, Chief Engineer and Chief Officer
Weather routeing	Operational Manager	Master
Voyage optimisation	Operational Manager	Master
Virtual arrival	Operational Manager	Master
Auto pilot controls	Operational Manager and Superintendents	Master

Compliance with each of these controls is verified during vessel visits such as audits and inspections.



SECTION 06

Control — Machinery and Power

Optimising on board power management, such as the use of main engine, generators and boilers, is one of the significant tools for energy saving on board.

06

Two places to look first

The methods that follow are to be used as far as they do not compromise safety, and for applicable vessels only.



Main engine

Standby time, constant speed running, load, load sharing, cylinder oil, turbocharger and exhaust boiler condition, performance monitoring and fuel treatment.



Auxiliary engines and boilers

Power demand per operation, unnecessary machinery, optimum loading, jet assist, air conditioning, tank cleaning, cargo heating and cargo tank pressure.

Ten practices that reduce main engine consumption

- All engines on board should be kept at standby condition for as little time as possible — this should be ensured by good communication and planning on board
- Engines should not run at constant speed for a long time, as this increases consumption
- Engines should be used at optimum load, considering the voyage and the weather
- Pitch and load should be monitored properly
- Immediate speed increases should be prevented as far as possible
- Load share between cylinder units should be monitored to ensure proper sharing
- Cylinder oil consumption should be monitored to ensure it is at the optimum level
- Turbocharger and exhaust gas boiler cleaning should be conducted properly
- Back pressure should be monitored
- Performance monitoring should be conducted, and fuel separation and treatment should be conducted properly



Monitoring is the control

Almost every item on this list is a monitoring duty. Nothing on the list requires new equipment — it requires attention, planning and honest records.

Running only the machinery you need



Determine the power each operation needs

Shipboard operations should be closely monitored and the required power determined for loading, discharging, ballasting, navigation, tank cleaning and every other operation.



No spare machinery on line

Additional auxiliaries or boilers should not be used unless necessary for the operation in hand.



Stop auxiliaries after the bow thruster

Additional auxiliaries should be stopped whenever it is certain that the bow thruster will not be used any more.



Load auxiliaries optimally

Auxiliaries should be used at optimum load, as per the manufacturer instructions.



Maintain the jet assist system

The jet assist system, if fitted, should be maintained in good condition. This should prevent the automatic start of another auxiliary during high demand.

Four more places energy is lost



Air conditioning and doors

Air conditioning should be used at optimum levels, and all accommodation entrances should be kept closed at all times.



Tank cleaning and cargo heating

These operations should be properly planned and monitored to ensure the minimum use of auxiliaries and boilers.



Cargo tank pressure

Pressure of cargo tanks should be properly monitored to prevent unnecessary use of machinery.



Stowage of heated cargoes

For heated cargoes a good stowage plan should be prepared. Cargoes requiring more heating should not be stowed in the forward part, as far as practicable.

Verification and responsibility for power management



How compliance is verified

Vessel visits

- Compliance with the requirements of this section is verified during vessel visits such as audits and inspections
- Machinery running hours, load and consumption records support the verification



Who is responsible

Office and vessel

- Office — Operational Manager and Superintendents
- Vessel — Master, Chief Engineer and Chief Officer



SECTION 07

Control — Hull, Propeller, Ballast

Condition of the underwater hull and propeller, the ballast plan and the equipment fitted all change how much fuel the ship burns to do the same work.

07

Propeller polishing and cleaning

Like the hull surface, propellers suffer degradation in performance due to surface roughness, which may be caused by corrosion, cavitation erosion and impingement attack, or by normal wear and tear. Improper maintenance can also increase roughness.

3%

Fuel saving that may result from polishing a roughened propeller surface

Dry dock

When polishing and cleaning are carried out, or whenever deemed necessary



What polishing recovers — Propeller polishing reduces mainly the frictional losses of the propeller, but also its rotational losses.



A cost effective measure — Cleaning and polishing of the propeller is a cost effective energy efficiency measure.



Driven by condition monitoring — Condition monitoring is conducted by vessel and office crew during ship visits, and determines when work is needed between dockings.

Hull cleaning



Fouling costs fuel

Hull fouling and roughness increase the fuel consumption of the ship. Hull cleaning reduces those losses.



Cost effective

Cleaning is a cost effective energy efficiency measure, not an optional cosmetic exercise.



When it is done

Cleaning is conducted on fleet vessels during dry dock periods, or whenever deemed necessary subject to condition monitoring by vessel and office crew during ship visits.

Why ballast operations matter to energy

Ballasting and deballasting operations at ports require extra generator and boiler use. The aim is to minimise the period of operations that draw on those additional energy sources.



Gravity always comes first

Priority will always be given to ballasting and deballasting by the gravity method, because it uses no pumping energy at all.



Then consider what else is running

When the operation reaches levels that cannot be achieved by gravity, the pump choice is made taking into account the other operations being carried out at that moment.



Time may override the preference

If there is a problem regarding time, then another pump can be used as well.

Choosing the ballast pump

The order below applies on ships where steam driven pump alternatives are fitted.

1

Gravity method

Always the first choice for ballasting and deballasting, wherever the levels allow it.

2

Steam ballast pump

Where steam driven pump alternatives are fitted and a double boiler is already in charge, the steam ballast pump is used.

3

Electrical ballast pump

Where no boiler is already in charge, the electrical ballast pump is taken instead.

4

Any other pump

If there is a problem regarding time, another pump can be used as well.

Four ballast practices to apply



Exchange and treatment

Where there is no problem with time and location, use gravity first, then the electric pump, and the steam pump if necessary and if fitted. Plan the operation together with other operations such as tank cleaning and purging where applicable.



Heavy weather ballasting

Heavy weather ballasting is not suggested unless necessary. If the Master decides on it from his own experience and foresight, office approval is necessary, and the precautions above still apply.



Minimum ballast amount

For Suezmax and Capesize ships a minimum ballast amount is determined that keeps the propeller fully immersed and balances the minimum draft limits. Ships are requested to apply it whenever weather allows.



Ballast stripping at ports

Stripping by eductor after pumping costs time and energy. At ports, where there is no draft limitation or risk of freezing, avoid leaving ballast to be stripped. The Master evaluates each case and strips only if necessary.

Two retrofits already in the fleet



Mewis Duct

A power-saving device

- A novel power-saving device developed for slower ships with a full form hull shape
- It allows either a significant fuel saving at a given speed, or a higher speed for a given power level
- It consists of two strong fixed elements — a duct positioned ahead of the propeller together with an integrated fin system
- Some applicable fleet vessels have been fitted with a Mewis Duct during their last dock period



Silicone anti-fouling paints

Foul release coatings

- A new generation of paints employing a foul release mechanism
- Foul release does not use biocides to control fouling
- It provides an ultra-smooth, slippery, low friction, hydrophobic or hydrophobic and hydrophilic combination surface on which fouling organisms have difficulty settling
- Foul release products contain no added biocides and are based on silicone and fluoro-polymer technology

Lighting management optimisation

The methods below are to be used for energy saving on board and in the office alike.



- Turn off lights when you finish work in a space
- Turn off the lights in spaces that are usually unoccupied — this includes the cabin WC
- Remove lamps that flicker without starting, because the cathodes are worn out
- Expect compliance to be checked during vessel visits, audits, inspections and office audits



- Leave lighting burning in empty accommodation, stores and working spaces
- Keep trying to start a flickering lamp — repeated attempts cause overheating and energy loss
- Treat lighting as too small to matter, on board or in the office

Building energy performance in at the design stage

The Company considers environmental and energy performance improvement opportunities and operational control in the design of new building vessels, and for modified equipment, systems and processes that can have a significant impact on performance.



Hull form optimisation

The shape of the hull is optimised for the intended service before the ship is built.



Energy saving devices

For example LEDs, and variable frequency drives on heavy power consumers.



VOC saving arrangements

Arrangements that reduce the loss of volatile organic compounds.



Clean fuel technology

Technology that allows cleaner fuels to be burned in service.



Waste reduction equipment

Equipment that reduces the waste generated by the ship in normal operation.



Recorded in the specification

The results of the energy and environmental performance evaluation are incorporated, where appropriate, into the specification, design and procurement activities of the relevant projects. The results of the design activity are recorded where appropriate.

THE LARGEST SINGLE MEASURE

Training and awareness of the crew is the greatest part of energy saving

The Company implements a training scheme for vessel and office crew regarding energy efficiency on board and in the office. Compliance is verified during vessel visits such as audits and inspections, and also during office audits.

Office — Operational Manager and Human Resources Manager

Vessel — Master, Chief Engineer and Chief Officer

Buying energy services, products and equipment



Today

Presently the Company does not procure any energy services.



When it does

When procuring energy services, products and equipment that have, or can have, an impact on significant energy use, the Company informs suppliers that procurement is partly evaluated on the basis of energy performance.



SECTION 08

Checking and Monitoring

The Company ensures that the key characteristics of its operations that determine energy performance are monitored continuously, measured and analysed.

What is monitored and reported from every ship

Energy Management includes monitoring and reporting requirements for the following on vessels.



Daily fuel consumption

Consumption of main engine, boilers, inert gas system and auxiliaries is followed by daily bunker monitoring and recorded on Bunker Reports.



Speed and distance travelled

Vessel speed and distance data is taken from the ECDIS, the log books and similar sources.



Vessel condition

Whether the vessel is laden or in ballast is taken from the log book.



Trim

Vessel trim is taken from the log book, and is also reported to the office with the noon reports.



Weather, sea state and wind

Weather, sea state and wind direction are taken from the log book.

Fleet data and office data



Fleet and vessel data

From the ships

- Data is recorded on a voyage by voyage basis
- Data is recorded for individual vessels and on an overall fleet basis
- Time spent alongside and at anchor is included in the data



Office consumption

Followed by official bills

- Electricity consumption
- Natural gas consumption
- Both are followed by the official bills received by the office

What else is monitored, measured and analysed

The other key characteristics of the Company monitored for Energy Management System purposes are as follows.

- Significant energy uses and the other outputs of the energy review
- The relevant variables related to significant energy uses
- Energy Performance Indicators
- The effectiveness of the action plans in achieving objectives and targets
- Evaluation of actual against expected energy consumption
- Compliance with the legal requirements and other requirements to which the Company subscribes, evaluated during the annual management review meetings



Who follows the results

The results from monitoring and measurement of the key characteristics are recorded and followed by the Operational Manager. Records of the compliance evaluations are maintained within the management review meeting reports.

How the system is audited

The Company conducts integrated internal audits every 12 months, on board and in the office, at planned intervals.

1

Planned and scheduled

During audit planning and scheduling the Company considers the status and importance of the processes and areas to be audited, as well as the results of previous audits.

2

Auditors selected for impartiality

The selection of auditors and the conduct of audits ensure the objectivity and impartiality of the audit process.

3

Conducted on board and ashore

The audits are integrated, covering the vessel and the office within the same programme.

4

Recorded and reported

Records of the audit results shall be maintained and reported to top management.

What the internal audit must confirm

The audit is conducted to ensure that the Environmental Management System meets the following.



Conforms to planned arrangements

It conforms to the planned arrangements for energy management, including the requirements of ISO 50001.



Conforms to objectives and targets

It conforms with the energy objectives and targets that have been established.



Works and improves

It is effectively implemented and maintained, and it improves energy performance.

Non-conformities, correction and corrective action

The Company addresses actual and potential non-conformities by making corrections and by taking corrective and preventive action.

1

Review

Review the non-conformities or potential non-conformities that have been raised.

2

Determine the causes

Determine the causes of the non-conformities or potential non-conformities.

3

Evaluate the need for action

Evaluate the need for action to ensure that non-conformities do not occur or recur.

4

Determine and implement action

Determine and implement the appropriate action needed, and make any necessary changes to the Energy Management System.

5

Maintain records

Maintain records of the corrective actions and preventive actions taken.

6

Review effectiveness

Review the effectiveness of the corrective action or preventive action taken. For reporting and analysis of non-conformities see M1 Main Manual Chapter 9.

Control of records and energy budget planning



Control of records

Legible and traceable

- Records must remain legible, identifiable and traceable to the relevant activity
- Records of monitoring, audits, corrective action and management review are all maintained



Energy budget planning

Before every voyage

- To avoid unnecessary bunker consumption, every vessel in the fleet should create an energy budget and planning form prior to commencement of her voyage
- The form should then be sent to the office
- At the end of the voyage the planning is compared with the real consumptions

The reporting sequence in the system

Vessels should fill in this section for every voyage. All data required by the system should be filled in correctly.

1

New voyage

Click the NEW VOYAGE button and enter the voyage start date and time. This will be the same as the COSP time on the departure report.

2

Noon reports

Fill in NOON REPORTS every day at noon by clicking the NEW NOON REPORT button. Noon reports should be filled in every day.

3

Arrival reports

At arrival port, canal transit, anchorage or drifting area, click NEW ARRIVAL REPORT. This form should be filled in at every arrival, at EOSP.

4

Departure reports

At departure from port, click NEW DEPARTURE REPORT. This form should be filled in at departure, at COSP.

5

Statement of facts

For every cargo or bunker call, canal transit, waiting at anchor or waiting adrift, fill in a SOF by clicking the NEW SOF button.

6

Close the voyage

Close the voyage by entering the date and end time of COSP from the last discharge port of the voyage.

How voyages are tracked in the system



The next voyage starts

The next voyage starts with a new departure report from the last discharge port of the previous voyage.



Closed voyages are locked

All closed voyages are shown as passive in the system and are locked against further changes.



Ongoing voyages are active

Voyages in progress are monitored as active voyages by the office.



SECTION 09

Management Review

During the annual management review meetings, the Energy Management System is reviewed to ensure its continuing suitability, adequacy and effectiveness.

What the management review is for



Suitability

That the system still fits the nature and scale of the organisation energy use and consumption.



Adequacy

That the resources, structure and controls are still sufficient for the system to work.



Effectiveness

That the system is actually delivering improved energy performance. Records of the review are maintained within the management review meeting report.

What is reviewed at the annual meeting

The following subjects are reviewed during the annual management review meeting regarding the Energy Management System.

- Follow-up actions from previous management reviews
- Review of the energy policy
- Review of energy performance and the related Energy Performance Indicators
- Results of the evaluation of compliance with legal requirements, and changes in legal and other requirements to which the organisation subscribes
- The extent to which the energy objectives and targets have been met
- Integrated audit results, including the Energy Management System
- The status of corrective actions and preventive actions
- Projected energy performance for the following period
- Recommendations for improvement, if any



One meeting, whole system

The energy review, the evaluation of legal compliance and the review of objectives all come together at this meeting. It is where the year is judged and the next year is set.

The decisions and actions the review produces

Outputs from the management review include any applicable decisions or actions related to the following.



Energy performance

Changes in the energy performance of the organisation.



The energy policy

Changes to the energy policy itself.



Performance indicators

Changes to the Energy Performance Indicators.



Objectives and targets

Changes to objectives, targets or other elements of the system, consistent with the commitment to continual improvement.



Resources

Changes to the allocation of resources.



Energy saved is emission avoided

Energy management is a structured process that runs from the energy policy through the operational controls on board to the annual management review. Every seafarer and every office employee has a defined part in it.



Monitor, record and report accurately



Run only the machinery you need



Optimise trim, speed and route



Suggest improvements you can see